



**FATIMA JINNAH WOMEN UNIVERSITY
RAWALPINDI**

Department of Software Engineering

SUBJECT: CLOUD COMPUTING LAB

Name: Saira Ejaz

Reg No: 2023-BSE-057

Section: 5-B

LAB NO 3:

Task 1 – Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)

Steps:

1. Open your repository on **GitHub** and edit the README.md file directly in the browser.

<https://github.com/23-22411-057-sudo/Lab3>

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner *

 23-22411-057-sudo ▾

Repository name *

Lab3

✓ Lab3 is available.

Great repository names are short and memorable. How about **didactic-journey**?

Description

Lab3 public repository of Cloud computing lab

45 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

 23-22411-057-sudo / Lab3

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

 Lab3 Public

 Pin  Watch 0  Fork 0  Star 0

main 1 Branch 0 Tags

Add file ▾

 Code ▾

 23-22411-057-sudo Initial commit 9bcd7e · now 1 Commit

README.md Initial commit now

README

Lab3

Lab3 public repository of Cloud computing lab

About

Lab3 public repository of Cloud computing lab

 Readme

 Activity

 0 stars

 0 watching

 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

```
Syed@DESKTOP-S50GK51 MINGW64 ~ (main)
$ cd /c/Users/Syed/Documents

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents (main)
$ git clone https://github.com/23-22411-057-sudo/Lab3.git
Cloning into 'Lab3'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

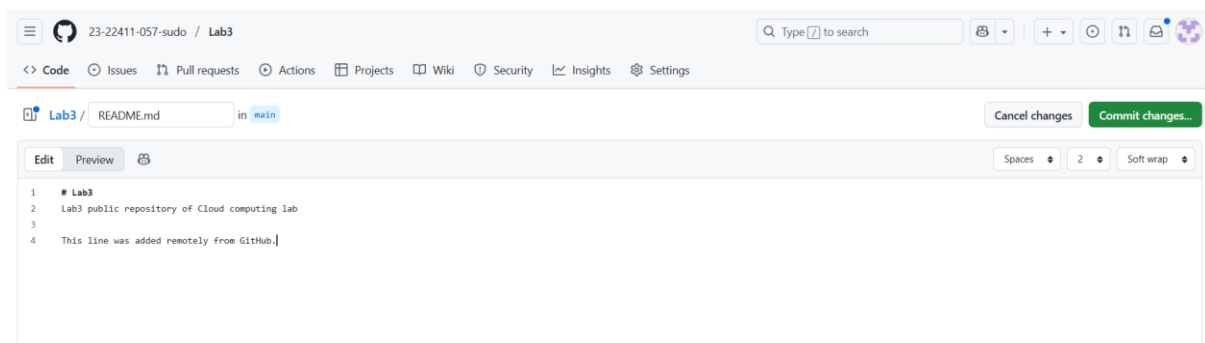
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents (main)
$ cd Lab3

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ ls
README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Add a new line such as:

This line was added remotely from GitHub.



Commit your change on GitHub (e.g., message: *Remote update to README*).

Save a screenshot of your browser showing the change as `remote_edit.png`.

Commit changes

Commit message

Remote update to README

Extended description

Add an optional extended description...

☒ Commit directly to the main branch

☐ Create a new branch for this commit and start a pull request [Learn more about pull requests](#)

Cancel

Commit changes

23-22411-057-sudo / Lab3

Type to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

main

Lab3 / README.md

23-22411-057-sudo

Remote update to README

Preview

Code

Blame

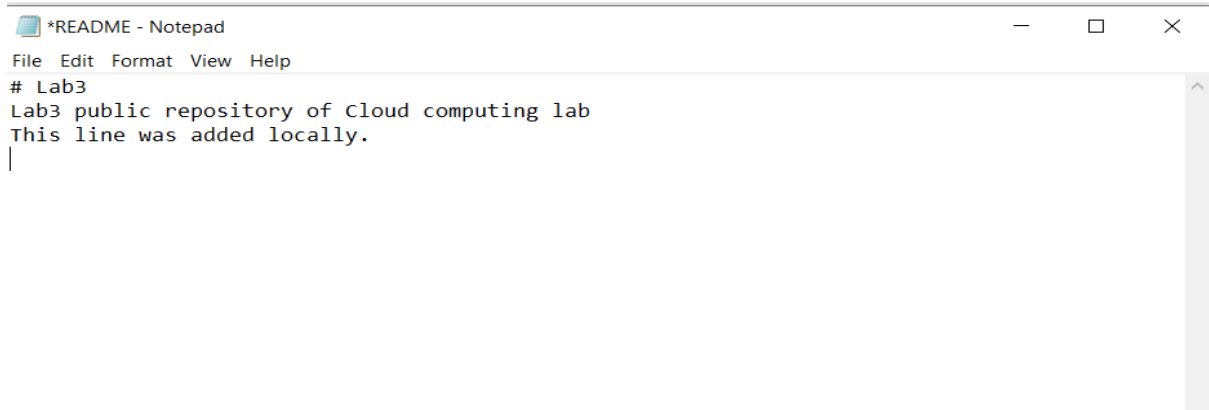
4 lines (3 loc) · 96 Bytes

Lab3

Lab3 public repository of Cloud computing lab

This line was added remotely from GitHub.

2. On your **local machine**, open the same repository folder.
 - a. Edit the same README.md file locally, for example:
This line was added locally.



3. Stage and commit your change:

git add README.md

git commit -m "Local update to README"

Save a screenshot of your terminal as local_commit.png.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Local update to README"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

4. Try to push:

git push origin main

You'll see an error message similar to:

hint: Updates were rejected because the remote contains work that you do not have locally.

Save a screenshot of this error as push_error.png.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push -u origin main
To https://github.com/23-22411-057-sudo/Lab3.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/23-22411-057-sudo/Lab3.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

5. To fix this, pull the latest changes from remote:

git pull --no-rebase origin main

Save a screenshot as merge_commit.png.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git pull --no-rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 987 bytes | 75.00 KiB/s, done.
From https://github.com/23-22411-057-sudo/Lab3
* branch      main      -> FETCH_HEAD
   9bcde7e..4f2a732  main      -> origin/main
Updating 9bcde7e..4f2a732
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Push again:

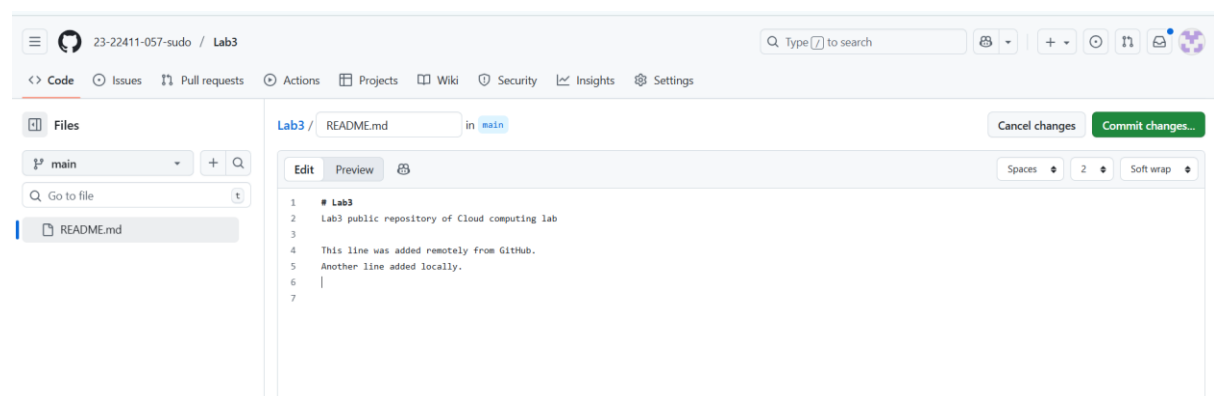
git push -u origin main

```
1 file changed, 2 insertions(+)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
```

6. Now repeat the same process, but this time use **rebase instead of merge**:

Make another remote edit on GitHub in README.md.



Then make another local edit and commit it.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Second local update"
[main fb835f4] Second local update
1 file changed, 2 insertions(+), 2 deletions(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

7. Instead of using git pull, run:

git pull --rebase origin main

```
1 file changed, 2 insertions(+), 2 deletions(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git pull --rebase origin main
From https://github.com/23-22411-057-sudo/Lab3
 * branch          main          -> FETCH_HEAD
Current branch main is up to date.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

8. Push again:

git push -u origin main

Have a screenshot as push_after_rebase.png.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 480 bytes | 240.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab3.git
   4f2a732..fb835f4  main -> main
branch 'main' set up to track 'origin/main'.

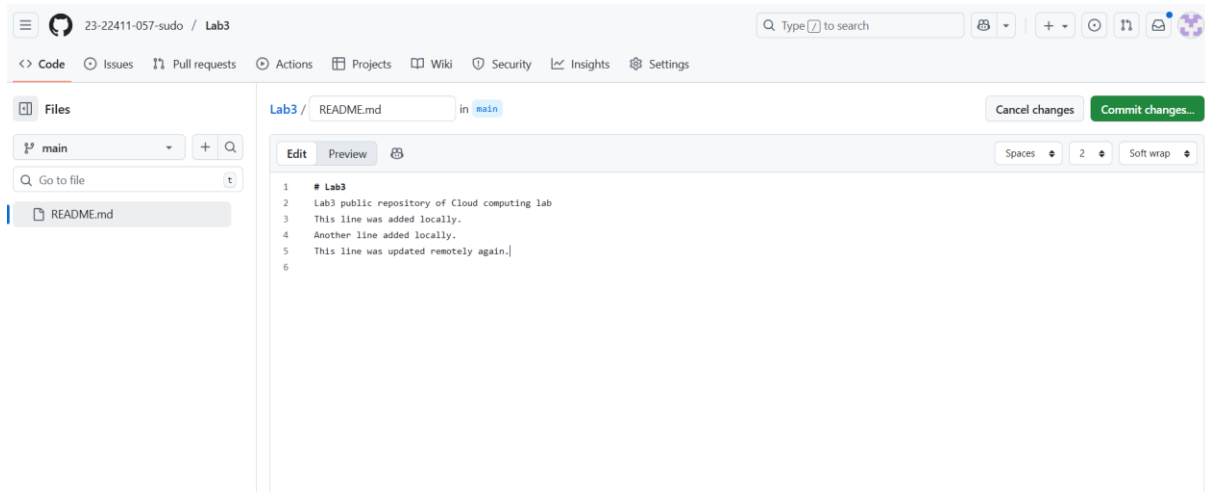
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Task 2 – Creating and Resolving Merge Conflicts Manually

This task will help you understand how Git handles file conflicts when two people modify the same line of code differently.

Steps

1. On **GitHub (remote)**, open your README.md file and change an existing line. Change it to: This line was updated remotely again.



Commit your change with the message: *Remote conflicting change*

Save a screenshot as remote_conflict_edit.png.

Commit changes

Commit message

Remote conflicting change

Extended description

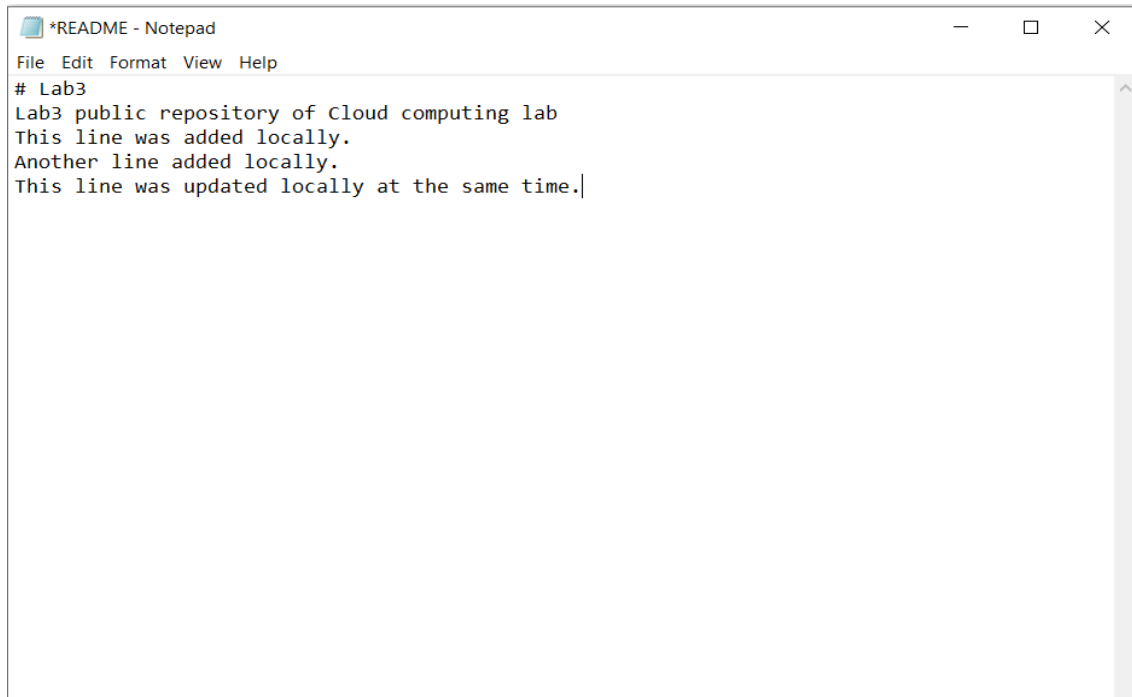
Add an optional extended description...

☒ Commit directly to the main branch

☐ Create a **new branch** for this commit and start a pull request [Learn more about pull requests](#)

Cancel Commit changes

2. On your **local machine**, edit the **same line** in the same file but make a **different change**:
3. This line was updated locally at the same time.



4. Stage and commit your local change:

git add README.md

git commit -m "Local conflicting change"

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Local conflicting change"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

5. Try to push:

git push -u origin main

The push will be **rejected**, since the remote version has conflicting changes.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push -u origin main
To https://github.com/23-22411-057-sudo/Lab3.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/23-22411-057-sudo/Lab3.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

6. Pull with rebase to bring in remote changes:

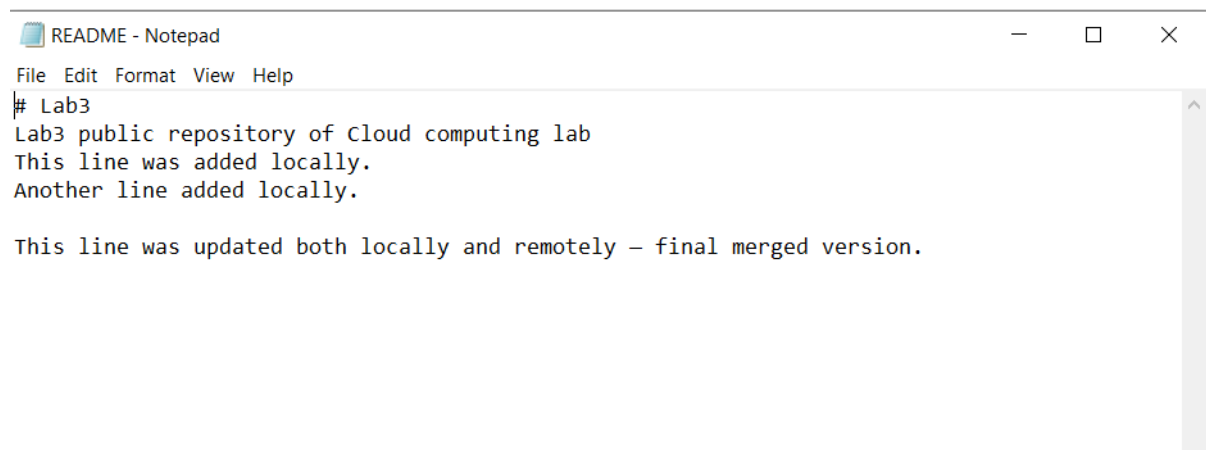
git pull --rebase origin main

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 965 bytes | 60.00 KiB/s, done.
From https://github.com/23-22411-057-sudo/Lab3
* branch          main      -> FETCH_HEAD
  fb835f4..6973dc5  main      -> origin/main
Updating fb835f4..6973dc5
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

7. Open the README.md file in your editor — you'll see conflict markers like this:

Manually edit the file to keep the correct line (for example, merge both ideas).



8. After fixing the file, mark the conflict as resolved and continue the rebase:

git add README.md

git rebase --continue

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git rebase --continue
fatal: no rebase in progress

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

9. Finally, push your changes:

git push -u origin main

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Task 3 – Managing Ignored Files with .gitignore and Removing Tracked Files

Steps

1. Create a new folder named textfiles inside your repository:

mkdir textfiles

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ mkdir textfiles
```

2. Inside the textfiles folder, create three text files:

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ echo "File A content" > textfiles/a.txt
echo "File B content" > textfiles/b.txt
echo "File C content" > textfiles/c.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ ls
README.md  textfiles/
```

3. Add and commit the new directory:

git add .

git commit -m "Added textfiles directory with three files"

git push origin main

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .
warning: in the working copy of 'textfiles/a.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'textfiles/b.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'textfiles/c.txt', LF will be replaced by CRLF the next time Git touches it

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Added textfiles directory with three files"
[main 0d14bfd] Added textfiles directory with three files
4 files changed, 5 insertions(+), 1 deletion(-)
create mode 100644 textfiles/a.txt
create mode 100644 textfiles/b.txt
create mode 100644 textfiles/c.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (7/7), 704 bytes | 176.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
6973dc5..0d14bfd  main -> main

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |

```

- Now, create a .gitignore file in the root of your repository:

```
echo "textfiles/*" > .gitignore
```

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ echo "textfiles/*" > .gitignore

```

- Add and commit the .gitignore file:

```
git add .gitignore
```

```
git commit -m "Added .gitignore to ignore textfiles directory"
```

```
git push origin main
```

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

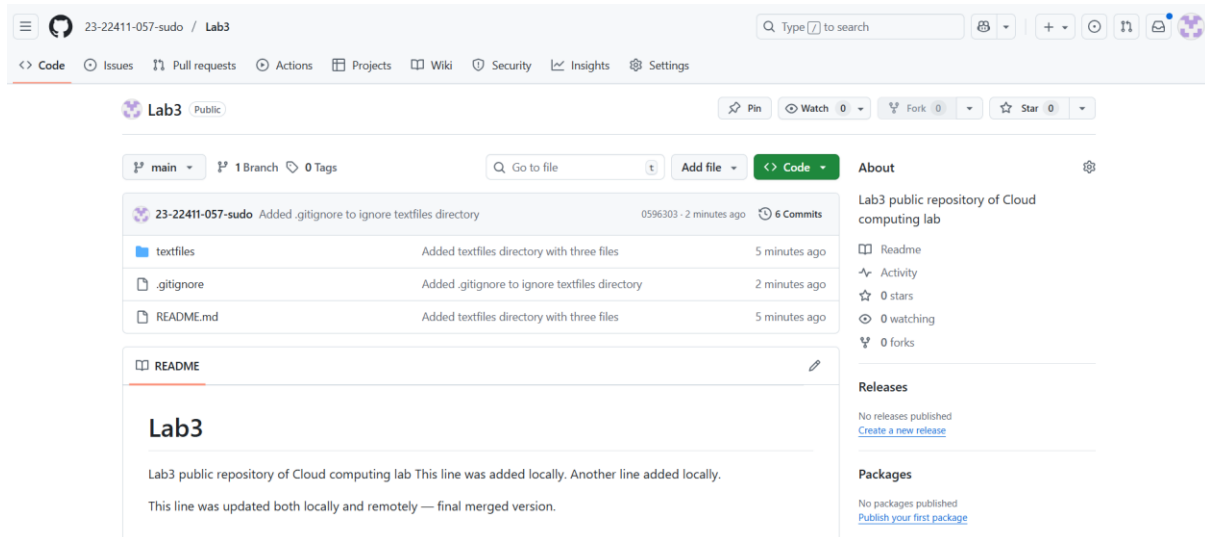
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Added .gitignore to ignore textfiles directory"
[main 0596303] Added .gitignore to ignore textfiles directory
1 file changed, 1 insertion(+)
create mode 100644 .gitignore

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 482 bytes | 241.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab3.git
0d14bfd..0596303  main -> main

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |

```

- Go to **GitHub** and check your repository — notice that the textfiles directory is **still visible** on the remote.



7. To remove already tracked files from Git (but not from your local system), run:

git rm -r --cached textfiles

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
```

8. Commit and push again:

git add .

git commit -m "Removed tracked textfiles directory"

git push origin main

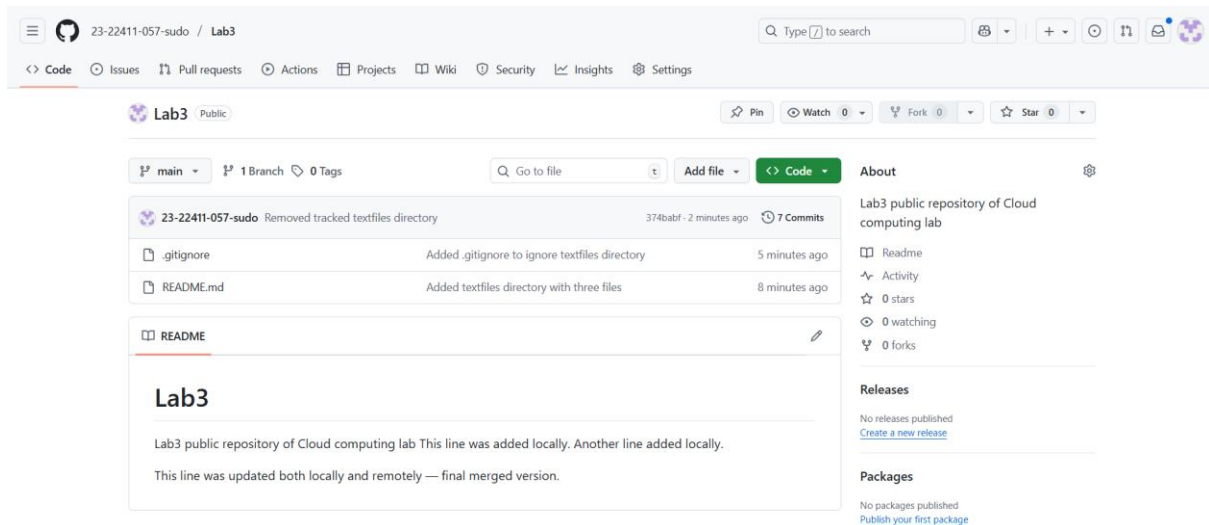
```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Removed tracked textfiles directory"
[main 374babf] Removed tracked textfiles directory
3 files changed, 3 deletions(-)
delete mode 100644 textfiles/a.txt
delete mode 100644 textfiles/b.txt
delete mode 100644 textfiles/c.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 359 bytes | 179.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
0596303..374babf main -> main

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

9. Check your GitHub repository again — the textfiles folder should now be **deleted remotely**.



Task 4 – Create Temporary Changes and Use git stash

Steps

1. Create a feature-branch.

git checkout -b feature-branch

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
```

2. Modify any file (for example, README.md) by adding a few test lines.

git add README.md

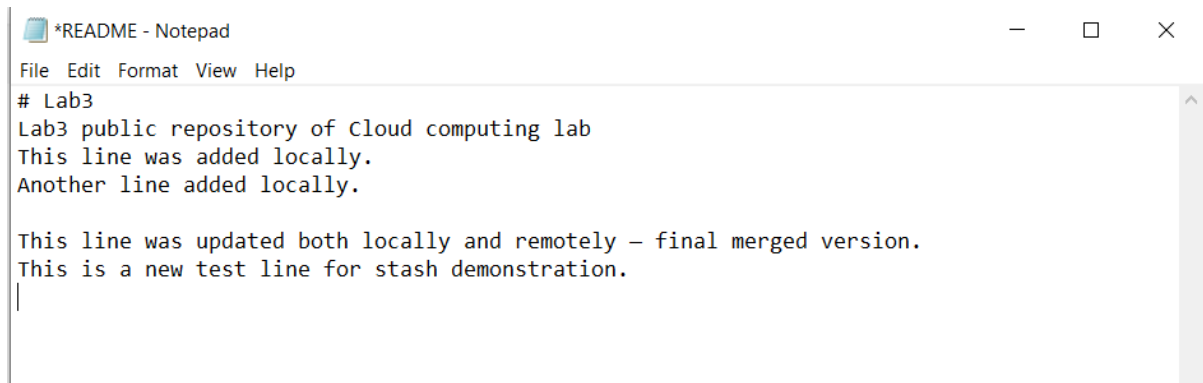
git commit -m "Changes the README.md file"

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git commit -m "Changes the README.md file"
On branch feature-branch
nothing to commit, working tree clean

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ |
```

3. Modify any file (for example, README.md) by adding a few new lines.

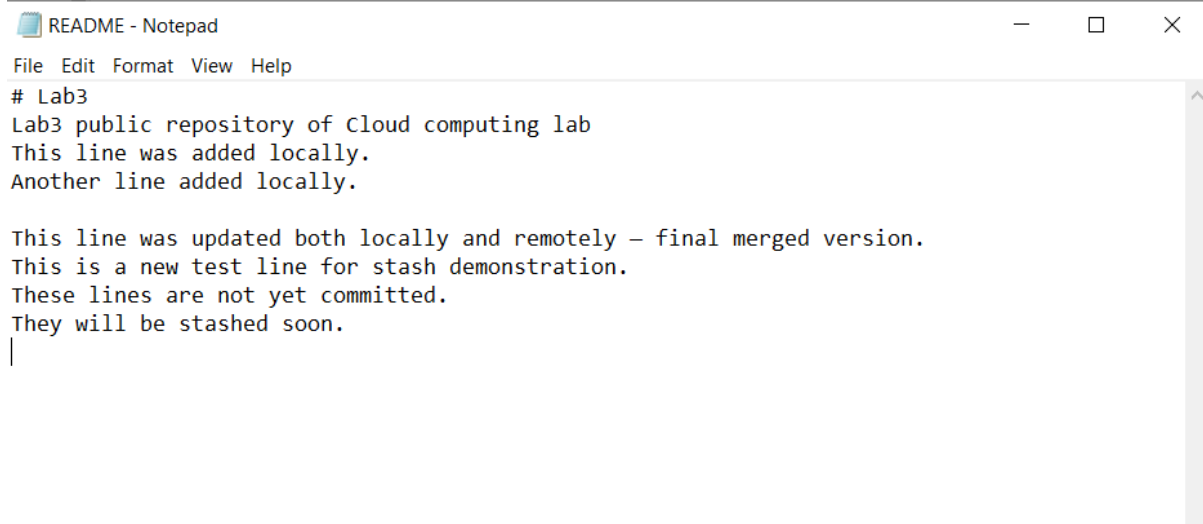


```
*README - Notepad
File Edit Format View Help
# Lab3
Lab3 public repository of Cloud computing lab
This line was added locally.
Another line added locally.

This line was updated both locally and remotely – final merged version.
This is a new test line for stash demonstration.
|
```

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git commit -m "Changes the README.md file"
On branch feature-branch
nothing to commit, working tree clean
```



```
README - Notepad
File Edit Format View Help
# Lab3
Lab3 public repository of Cloud computing lab
This line was added locally.
Another line added locally.

This line was updated both locally and remotely – final merged version.
This is a new test line for stash demonstration.
These lines are not yet committed.
They will be stashed soon.
|
```

4. Without committing or stashing the changes, try to switch to another branch:

git checkout main

You'll see an error message similar to:

Please commit your changes or stash them before you switch branches.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git checkout main
M      README.md
Already on 'main'
Your branch is up to date with 'origin/main'.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ This is the main branch version.
bash: This: command not found

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Updated README.md on main branch"
[main f768264] Updated README.md on main branch
 1 file changed, 4 insertions(+)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 750 bytes | 187.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab3.git
   374babf..f768264  main -> main

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git checkout feature-branch
Switched to branch 'feature-branch'

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
      README.md
Please commit your changes or stash them before you switch branches.
Aborting

```

5. To fix this, **stash your changes**:

git stash

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git stash
Saved working directory and index state WIP on feature-branch: 374babf Removed tracked textfiles directory

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ |

```

6. Try switching branches again — it should now work:

git checkout main

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

```

7. Return to your previous branch (for example, feature-branch):

git checkout feature-branch

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git checkout feature-branch
Switched to branch 'feature-branch'

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ |

```

8. Check your working directory status:

git status

It should show: nothing to commit, working tree clean

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git status
On branch feature-branch
nothing to commit, working tree clean
```

9. Now restore your stashed changes:

git stash pop

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (123f0d730d9e035d1de37b0ec50233b06844faf3)
```

Task 5 – Checkout a Specific Commit Using git log

1. View commit history:

git log --oneline

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git log --oneline
374babf (HEAD -> feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit
```

2. Copy any previous commit hash (e.g., a1b2c3d).

3. Checkout that commit:

git checkout <commit-hash>

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (feature-branch)
$ git checkout 0596303
M      README.md
Note: switching to '0596303'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 0596303 Added .gitignore to ignore textfiles directory

```

4. To return to your main branch:

git checkout main

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 ((0596303...))
$ git stash
Saved working directory and index state WIP on (no branch): 0596303 Added .gitignore to ignore textfiles directory

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 ((0596303...))
$ git checkout main
Previous HEAD position was 0596303 Added .gitignore to ignore textfiles directory
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git stash pop
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
On branch main
Your branch is up to date with 'origin/main'.

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
The stash entry is kept in case you need it again.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |

```

Task 6 – Resetting Commits (Soft vs Hard Reset) (With Verification Steps)

Steps

1. Add a new line in any file and commit it:
 - # Edit any file (e.g., README.md), add a line
 - git add .
 - git commit -m "Added test line"

```

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Added test line"
[main e431084] Added test line
1 file changed, 2 insertions(+)

```

2. Add another change and commit again:

Edit the file again, add a different line

git add .

git commit -m "Second test commit"

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Second test commit"
[main 90967a5] Second test commit
1 file changed, 2 insertions(+), 1 deletion(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

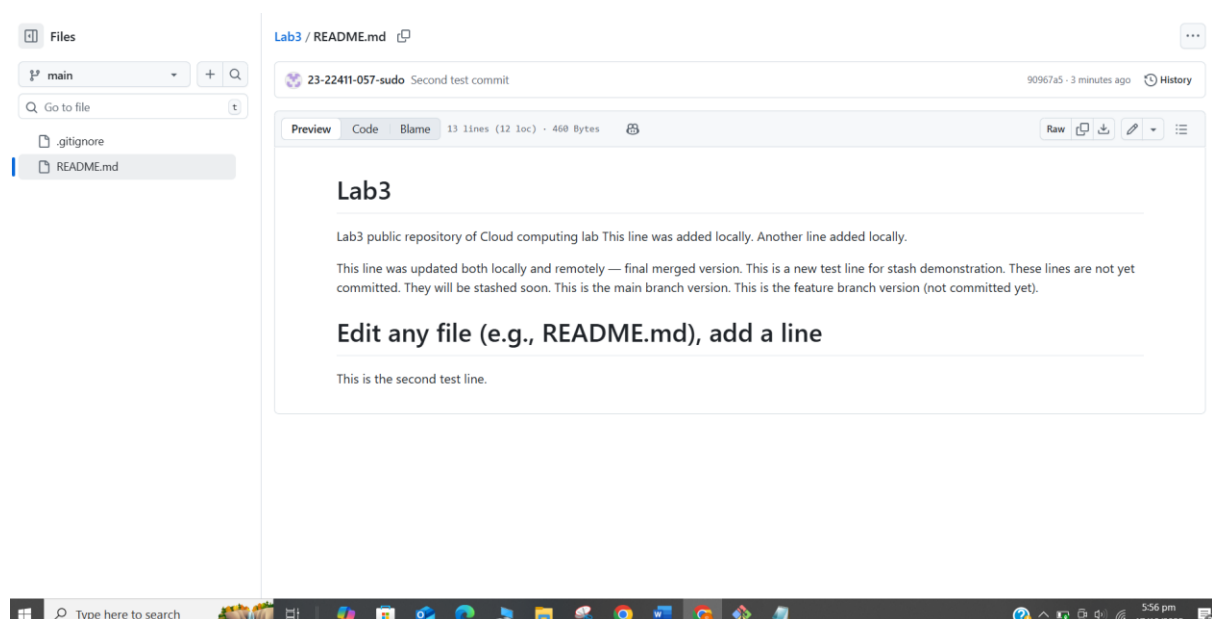
3. View the history before reset:

git log --oneline

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git log --oneline
90967a5 (HEAD -> main) Second test commit
e431084 Added test line
f768264 (origin/main, origin/HEAD) Updated README.md on main branch
374babf (feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit
```

4. Check file contents for both changes:

Open the edited file (e.g., README.md) and visually confirm BOTH added lines exist.



5. Perform a soft reset (keeps changes in working directory)

git reset --soft HEAD~1

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git reset --soft HEAD~1
```

6. Check commit history after soft reset

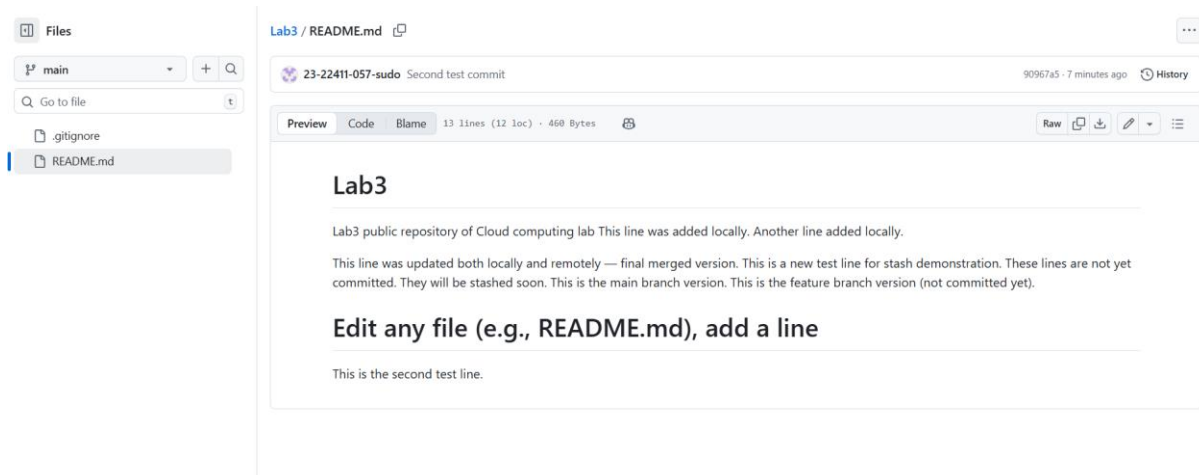
git log --oneline

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git log --oneline
e431084 (HEAD -> main) Added test line
f768264 Updated README.md on main branch
374babf (feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

7. Verify changes in the file:

Open the file again and confirm both edits are still present (nothing lost).



8. Check git status

git status

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
(use "git pull" to update your local branch)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

9. Perform commit

git commit -m "Second Test commit"

- Save a screenshot as file_after_hard_reset.png.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Second Test commit"
[main 7543ec1] Second Test commit
1 file changed, 2 insertions(+), 1 deletion(-)
```

10. Perform a hard reset (discards changes completely)

git reset --hard HEAD~1

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git reset --hard HEAD~1
HEAD is now at e431084 Added test line
```

11. Check commit history after hard reset

git log --oneline

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git log --oneline
e431084 (HEAD -> main) Added test line
f768264 Updated README.md on main branch
374babf (feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

12. Verify changes in the file:

Open the file and confirm the **second edit is gone** (reverted to previous commit).

```
README - Notepad
File Edit Format View Help
# Lab3
Lab3 public repository of Cloud computing lab
This line was added locally.
Another line added locally.

This line was updated both locally and remotely – final merged version.
This is a new test line for stash demonstration.
These lines are not yet committed.
They will be stashed soon.
This is the main branch version.
This is the feature branch version (not committed yet).
# Edit any file (e.g., README.md), add a line
```

13. Check git status

git status

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Task 7 – Amending the Last Commit

1. Make a small change in any file.

```
This is the feature branch version (not committed yet).
This is line one
This line was added for amend task.
```

2. Stage it and commit:

git add .

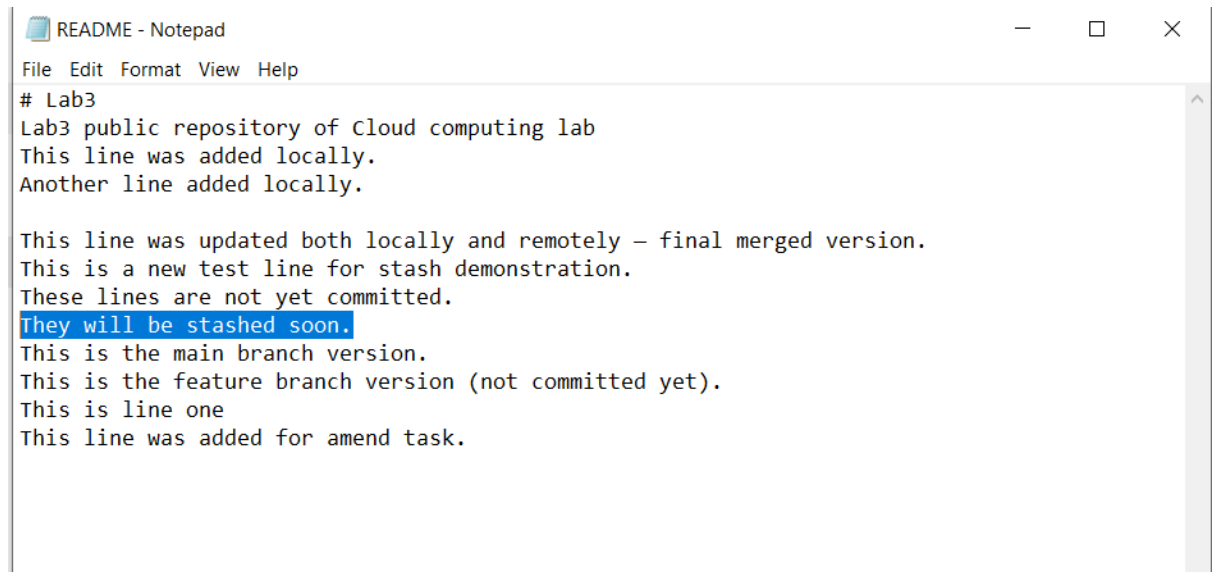
git commit -m "Fix log message"

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Fix log message"
[main 14674dd] Fix log message
1 file changed, 2 insertions(+), 1 deletion(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

3. Realize you forgot to change another file. Update it now.

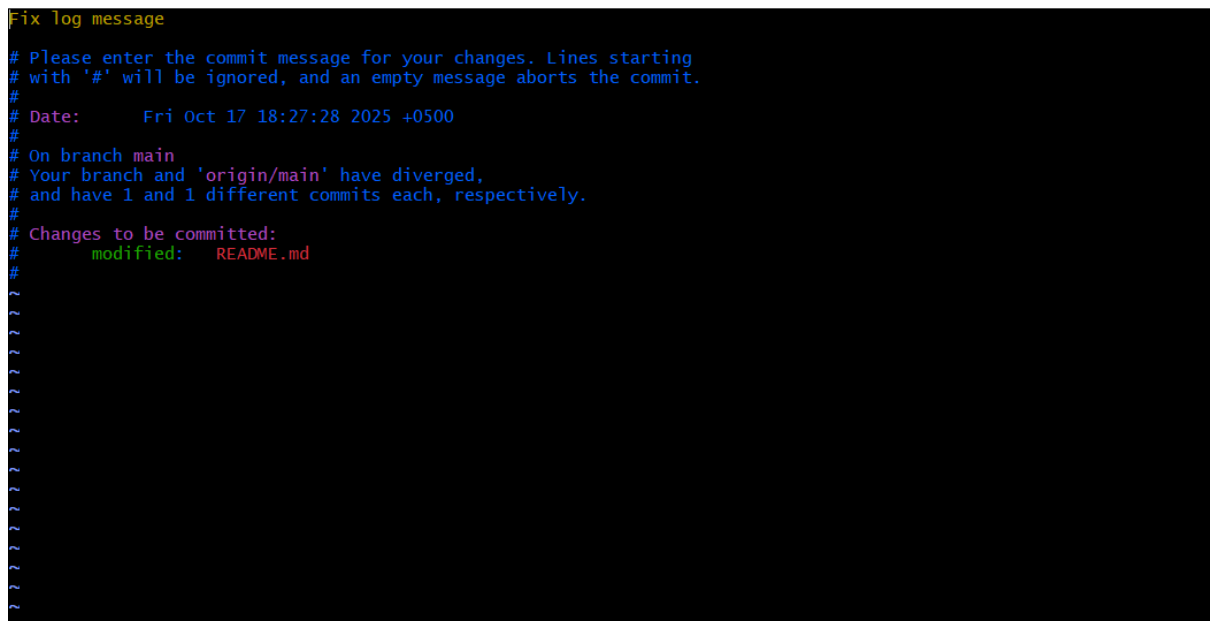


4. Add it and amend the last commit:

git add .

git commit --amend

- Save a screenshot as amend_commit.png.



```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git log --oneline
fa09b59 (HEAD -> main) Fix log message
e431084 Added test line
f768264 Updated README.md on main branch
374babf (feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit
```

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main --force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 489 bytes | 244.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
+ 90967a5...fa09b59 main -> main (forced update)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Task 8 – Reverting a Commit (Safe Undo on Remote Branch)

1. Make a change and commit it:

```
echo "temporary text" >> temp.txt
```

```
git add .
```

```
git commit -m "Added temporary text file"
```

```
git push origin main
```

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ echo "temporary text" >> temp.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git add .
warning: in the working copy of 'temp.txt', LF will be replaced by CRLF the next time Git touches it

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git commit -m "Added temporary text file"
[main 625ec04] Added temporary text file
1 file changed, 1 insertion(+)
create mode 100644 temp.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 464 bytes | 232.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab3.git
fa09b59..625ec04 main -> main

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

2. Now revert that commit safely (do not delete it):

```
git log --oneline
```


git revert <commit-hash>

```
Revert "Fix log message"
This reverts commit fa09b59c90d92b5c829935d1decf4655da37f244.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch main
# Your branch is up to date with 'origin/main'.
#
# Changes to be committed:
#   modified:   README.md
#
~
~
~
~
~
~
~
~
~
~
~
```

- ### 3. Push the revert commit:

```
git push origin main
```

```
[main 1015e60] Revert "Fix log message"
1 file changed, 2 insertions(+), 3 deletions(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 616 bytes | 123.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
   625ec04..1015e60  main -> main

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ |
```

Task 9 – Force Push (With Caution)

1. Create a new branch:

```
git checkout -b test-force
```

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (main)
$ git checkout -b test-force
Switched to a new branch 'test-force'
```

1. Make and commit a small change.

```
This is a test line for the force push task.
```

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ git add .

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ git commit -m "Added test line for force push demo"
[test-force 4ef3860] Added test line for force push demo
1 file changed, 6 insertions(+), 2 deletions(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ |
```

2. Push it:

git push origin test-force

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ git push origin test-force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 592 bytes | 296.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:   https://github.com/23-22411-057-sudo/Lab3/pull/new/test-force
remote:
To https://github.com/23-22411-057-sudo/Lab3.git
 * [new branch]      test-force -> test-force
```

3. Perform a hard reset:

git reset --hard HEAD~1

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ git reset --hard HEAD~1
HEAD is now at 1015e60 Revert "Fix log message"
```

4. Push again

git push origin test-force

Rejected by remote repository

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ git push origin test-force
To https://github.com/23-22411-057-sudo/Lab3.git
! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'https://github.com/23-22411-057-sudo/Lab3.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

5. Now force-push to remote:

git push origin test-force --force

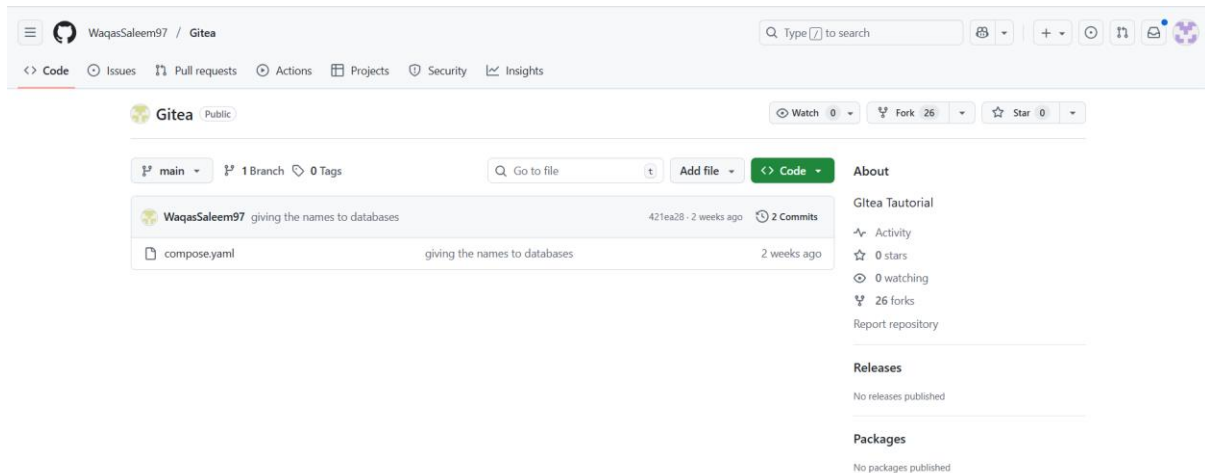
```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/Lab3 (test-force)
$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab3.git
+ 4ef3860...1015e60 test-force -> test-force (forced update)
```

Task 10 – Running Gitea in GitHub Codespaces via Docker Compose

Steps

1. Fork the Gitea Repository

Go to [Gitea](https://gitea.io).



Click the Fork button at the top right and fork it into your own GitHub account.

Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (*).

Owner *

 23-22411-057-sudo ▾

Repository name *

Gitea

✔ Gitea is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

Gltea Tautorial

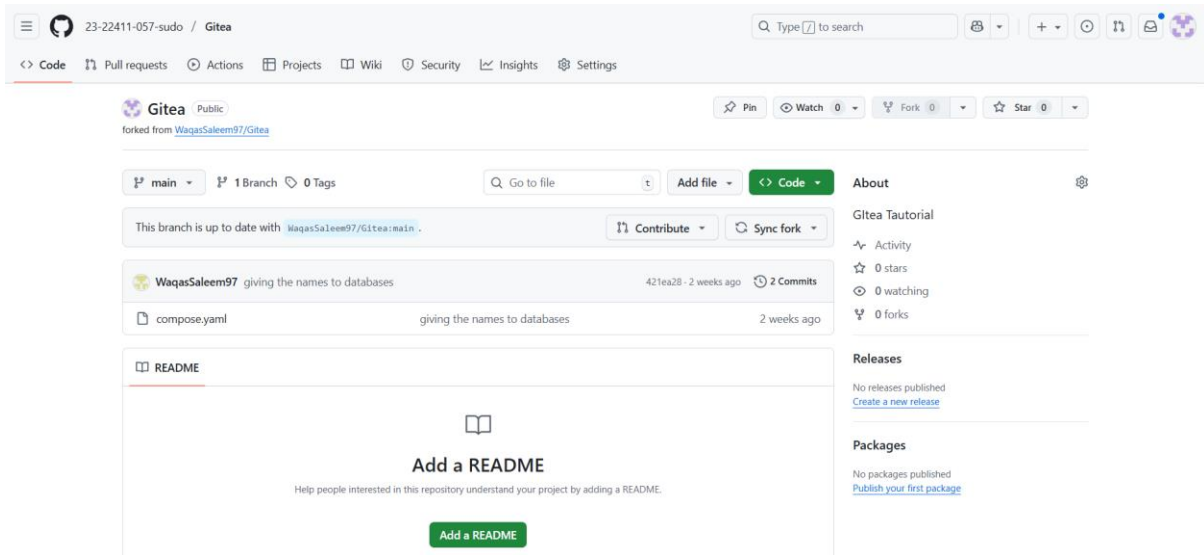
15 / 350 characters

☒ Copy the `main` branch only

Contribute back to WaqasSaleem97/Gitea by adding your own branch. [Learn more.](#)

 You are creating a fork in your personal account.

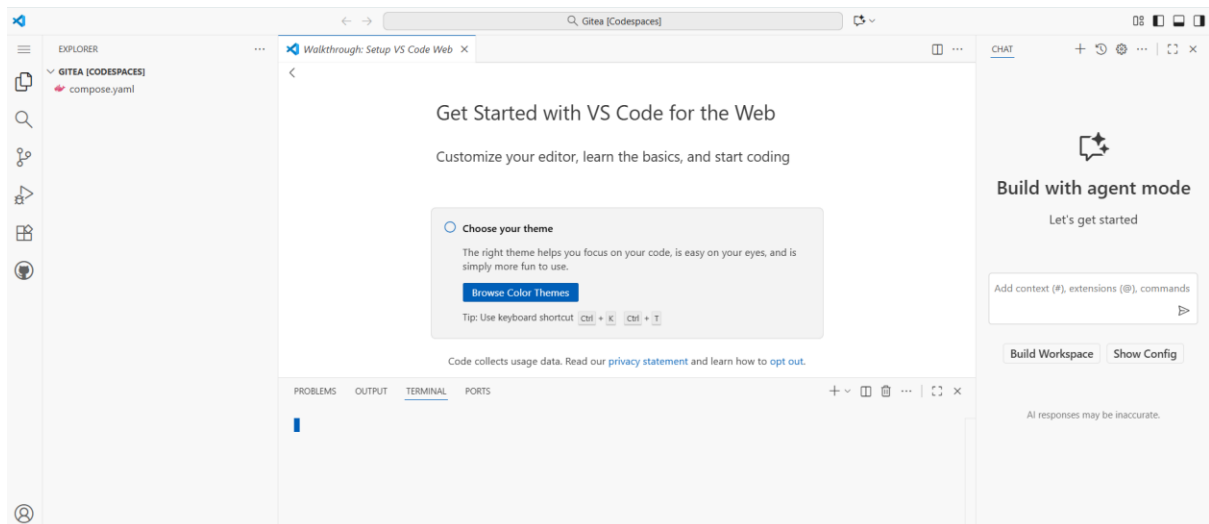
Create fork



The screenshot shows the Gitea web interface for a repository named 'Gitea' owned by '23-22411-057-sudo'. The repository is a fork of 'WaqasSaleem97/Gitea'. The main branch is 'main', and it is up to date with the upstream repository. The repository has 1 branch and 0 tags. A commit by 'WaqasSaleem97' is shown, titled 'giving the names to databases', dated 421ea28 - 2 weeks ago, with 2 commits. The repository has a README file. The right sidebar shows repository statistics: 0 stars, 0 watching, and 0 forks. There are sections for Releases and Packages, both indicating no published items.

2. Open the Forked Repo in GitHub Codespaces

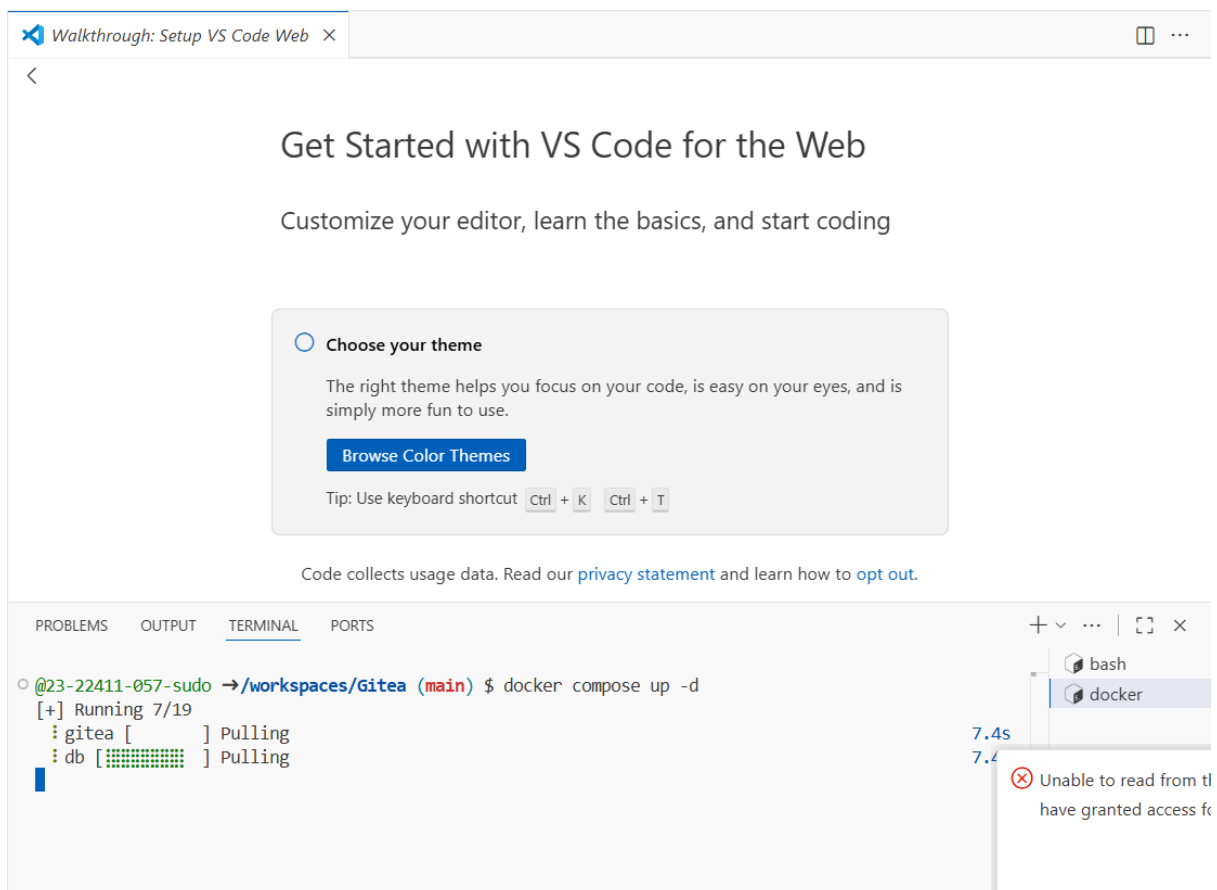
On your forked repo, click Code > Codespaces > Create codespace on main.

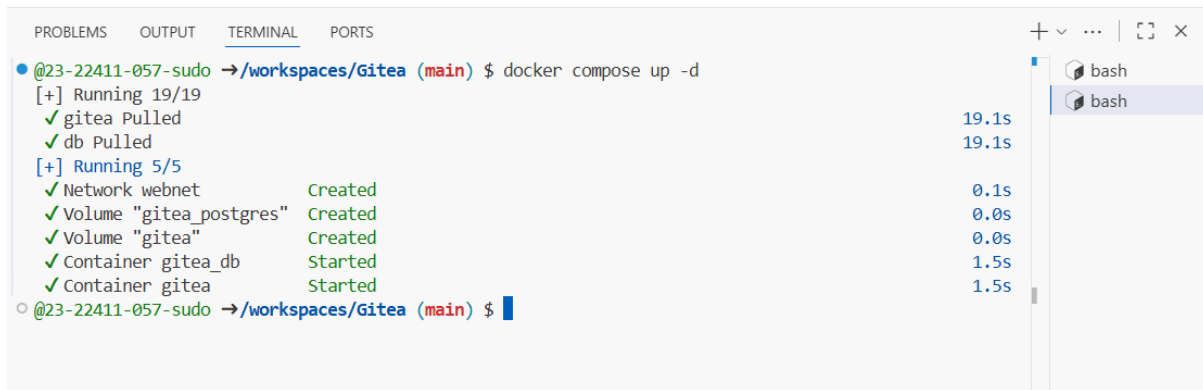


3. Start Gitea with Docker Compose

In the Codespace terminal, run:

docker compose up -d





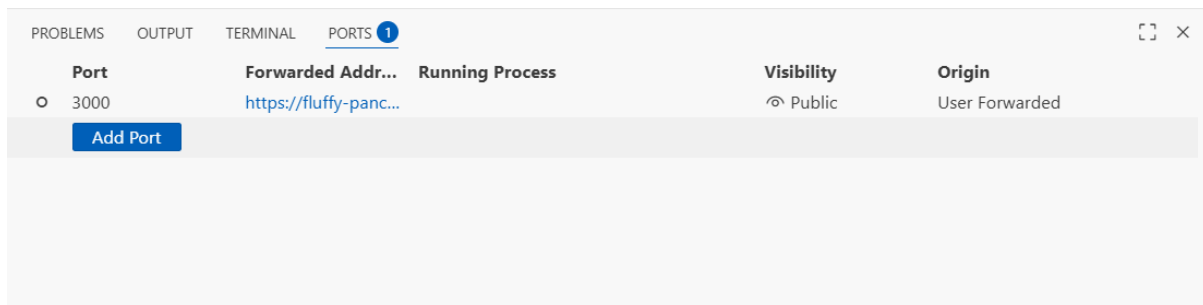
```
@23-22411-057-sudo →/workspaces/Gitea (main) $ docker compose up -d
[+] Running 19/19
 ✓ gitea Pulled
 ✓ db Pulled
[+] Running 5/5
 ✓ Network webnet      Created
 ✓ Volume "gitea_postgres" Created
 ✓ Volume "gitea"      Created
 ✓ Container gitea_db   Started
 ✓ Container gitea      Started
@23-22411-057-sudo →/workspaces/Gitea (main) $
```

4. Access Gitea Web Interface

In Codespaces, forward port 3000 (see Codespaces port forwarding UI).

Change the Visibility of port 3000 from private to public.

Click the forwarded port 3000 link to open Gitea in your browser tab.



Port	Forwarded Addr...	Running Process	Visibility	Origin
3000	https://fluffy-panc...		Public	User Forwarded

Add Port

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *

SQLite3

Path *

/data/gitea/gitea.db

File path for the SQLite3 database.

Enter an absolute path if you run Gitea as a service.

General Settings

Site Title *

Gitea: Git with a cup of tea

You can enter your company name here.

Repository Root Path *

/data/git/repositories

Remote Git repositories will be saved to this directory.

Git LFS Root Path

/data/git/lfs

Files tracked by Git LFS will be stored in this directory. Leave empty to disable.

Run As Username *

git

The operating system username that Gitea runs as. Note that this user must have access to the repository root path.

Server Domain *

fluffy-pancake-69x9r4r4jx4w3j4p-3000.app.github.dev

Domain or host address for the server.

SSH Server Port

22

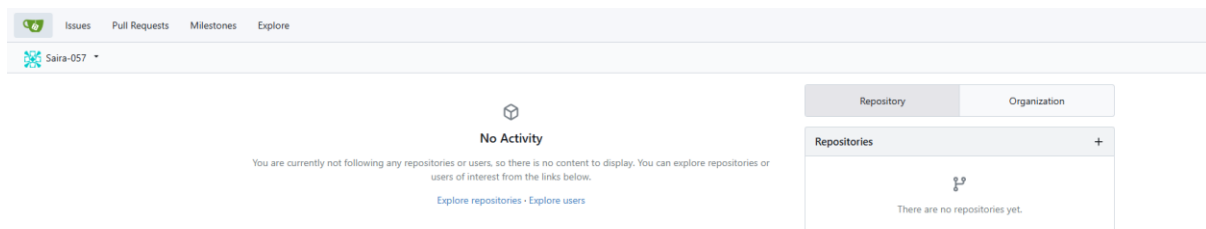
Port number your SSH server listens on. Leave empty to disable.

Gitea HTTP Listen Port *

3000

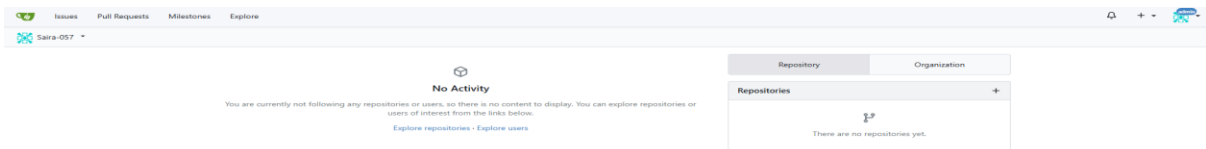
5. Install Gitea

Fill out the installation form, providing an adminusername and password (e.g., admin / yourpassword).



6. Log In to Gitea

Use your admin credentials to log in.



7. Create a New Repository in Gitea

Click New Repository or + > New Repository in Gitea.

Fill out the repository details and create it.

Save a screenshot of the new repository page as gitea_new_repo.png.

New Repository

A repository contains all project files, including revision history. Already hosting one elsewhere? [Migrate repository.](#)

Owner *

 Saira-057

Some organizations may not show up in the dropdown due to a maximum repository count limit.

Repository Name *

Lab3_Gitea_Test

Good repository names use short, memorable and unique keywords. A repository named ".profile" or ".profile-private" could be used to add a README.md for the user/organization profile.

Visibility

☐ Make repository private

Only the owner or the organization members if they have rights, will be able to see it.

Description

Sample Repository

Template

Select a template.

Issue Labels

Select an issue label set.

.gitignore

Select .gitignore templates.

Choose which files not to track from a list of templates for common languages. Typical artifacts generated by each language's build tools are included on .gitignore by default.

License

Select a license file.

A license governs what others can and can't do with your code. Not sure which one is right for your project? See [Choose a license.](#)

README

Default

The image shows the 'Create Repository' form on GitHub. It includes several sections: a '.gitignore' section with a dropdown menu and explanatory text; a 'License' section with a dropdown menu and explanatory text; a 'README' section with a dropdown menu and explanatory text; a checkbox for 'Initialize Repository (Adds .gitignore, License and README)'; a 'Default Branch' section with a dropdown menu and explanatory text; an 'Object Format' section with a dropdown menu and explanatory text; and a 'Template' section with a checkbox. At the bottom is a blue 'Create Repository' button.

.gitignore Select .gitignore templates.
Choose which files not to track from a list of templates for common languages. Typical artifacts generated by each language's build tools are included on .gitignore by default.

License Select a license file.
A license governs what others can and can't do with your code. Not sure which one is right for your project? See [Choose a license](#).

README Default
This is the place where you can write a complete description for your project.

☒ Initialize Repository (Adds .gitignore, License and README)

Default Branch main
The default branch is the base branch for pull requests and code commits.

Object Format sha1
Object format of the repository. Cannot be changed later. SHA1 is most compatible.

Template ☐ Make repository a template

Create Repository

Task 11 – Creating a GitHub Pages Portfolio Site

This task will guide you through creating a personal portfolio/static site using GitHub Pages.

Steps

1. Create a GitHub Pages Repository

Go to GitHub and create a new repository named <your-username>.github.io (e.g., if your username is WaqasSaleem97, the repo should be WaqasSaleem97.github.io).

Make sure the repository is **public**.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *

 23-22411-057-sudo ▾

Repository name *

/ 23-22411-057-sudo.githubs.io

✓ 23-22411-057-sudo.githubs.io is available.

Great repository names are short and memorable. How about **cautious-fortnight**?

Description

Repository for Portfolio

24 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

Create repository

2. Add Static Website Code

Prepare a simple static website (e.g., your CV or portfolio) in HTML/CSS/JS by using the sample code provided in the repository [Iqra Bashir CV](#).

Put your files in a local folder.

Documents > SPORTFOLIO

☐ Name	Date modified	Type	Size
index (2)	17/10/2025 7:23 pm	Chrome HTML Do...	5 KB
style	17/10/2025 7:21 pm	Cascading Style Sh...	2 KB

3. Push the Files to GitHub

Initialize git in your folder (if not already):

```
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ git remote remove origin

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ git remote add origin https://github.com/23-22411-057-sudo/23-22411-057-sudo.githubs.io.git

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ git branch -M main

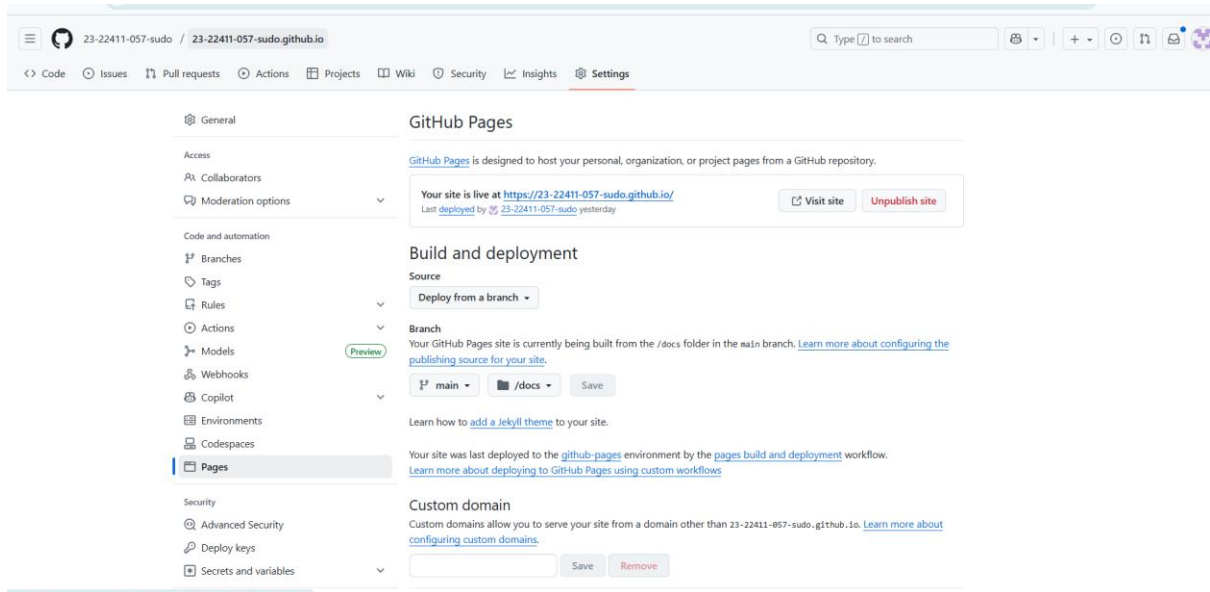
Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ git add .

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ git commit -m "Add portfolio site for GitHub Pages"
On branch main
nothing to commit, working tree clean

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 6.77 KiB | 1.35 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/23-22411-057-sudo.githubs.io.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Syed@DESKTOP-S50GK51 MINGW64 ~/Documents/SPORTFOLIO (main)
$ |
```

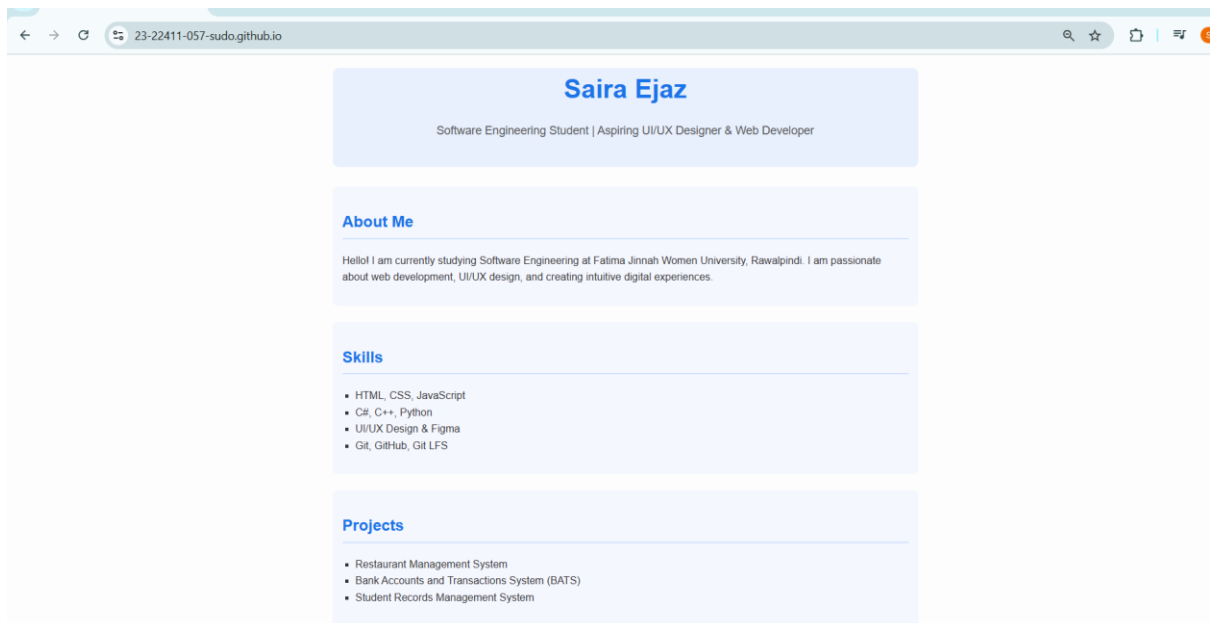
4. Check GitHub Pages Settings



5. Visit Your Live Site

Open your GitHub Pages site in the browser.

<https://23-22411-057-sudo.github.io/2023-BSE-057-sudo.github.io/>

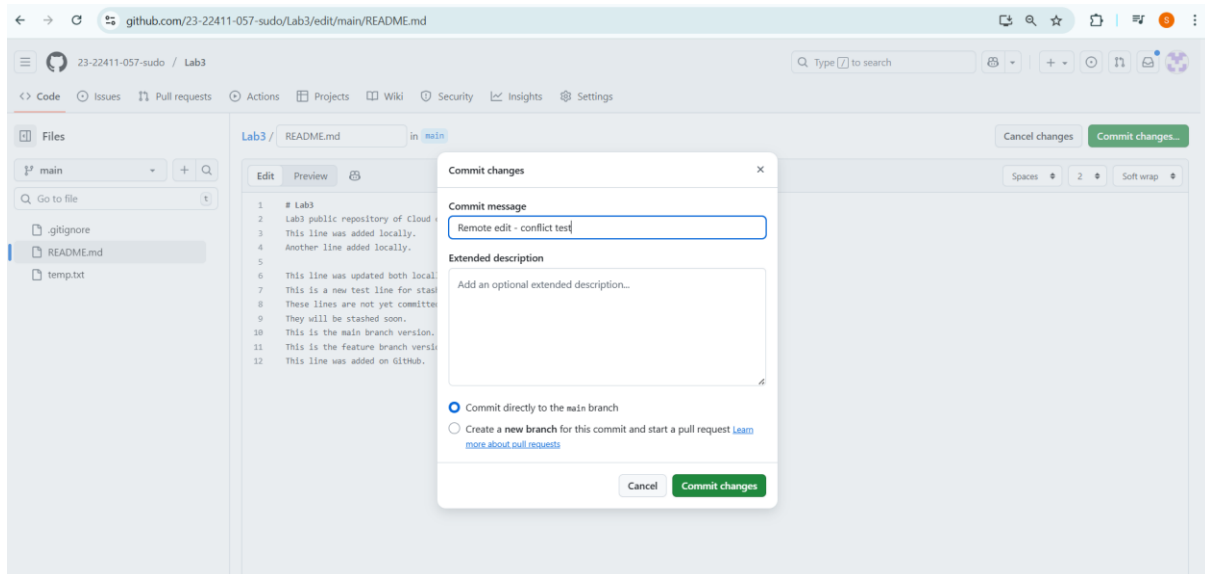


Exam Evaluation Questions

1. Local vs Remote Conflict Resolution

Steps:

1. On GitHub, edit a file (e.g., README.md) and commit the change.



2. On your local machine, edit the same file differently (Avoid Conflict) and commit.

```
Syed@DESKTOP-S50GK51 MINGW64 ~ (main)
$ cd documents

Syed@DESKTOP-S50GK51 MINGW64 ~/documents (main)
$ cd Lab3

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (test-force)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (test-force)
$ git commit -m "Local edit - conflict test"
[test-force 9ff77a9] Local edit - conflict test
1 file changed, 6 insertions(+), 2 deletions(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (test-force)
$
```

3. Try to push your local commit and observe the error.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
To https://github.com/23-22411-057-sudo/Lab3.git
! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/23-22411-057-sudo/Lab3.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$
```

4. Resolve the conflict by running git pull (merge), then push.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.00 KiB | 44.00 KiB/s, done.
From https://github.com/23-22411-057-sudo/Lab3
* branch          main          -> FETCH_HEAD
   1015e60..ad9c822  main          -> origin/main
Updating 1015e60..ad9c822
Fast-forward
 README.md | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

5. Repeat with another remote/local change, but resolve using `git pull --rebase`, then push.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "Resolved merge conflict"
[main a734e00] Resolved merge conflict
 1 file changed, 6 insertions(+), 2 deletions(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 570 bytes | 285.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
   ad9c822..a734e00  main -> main

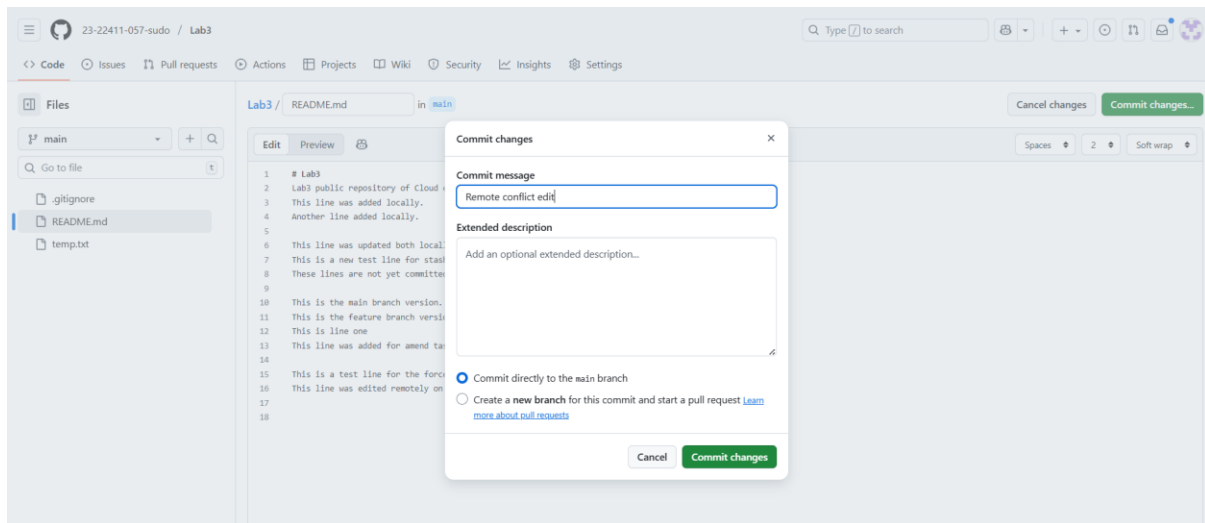
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

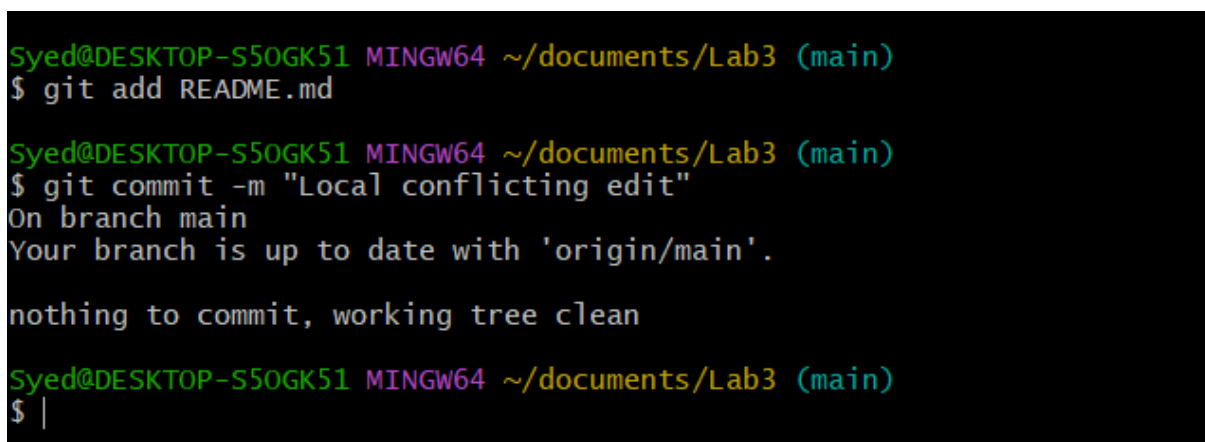
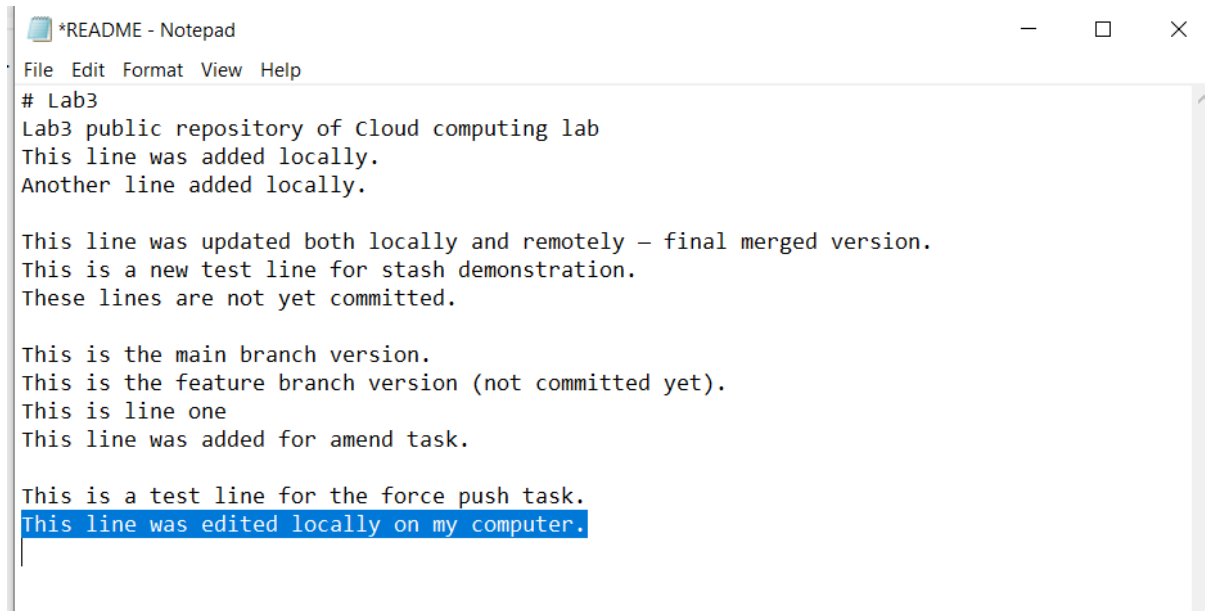
2. Manual Merge Conflict Handling

Steps:

1. On GitHub, change a specific line in a file and commit.



2. Locally, change the same line differently and commit.



3. Try to push your local change and observe the conflict error.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
To https://github.com/23-22411-057-sudo/Lab3.git
! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/23-22411-057-sudo/Lab3.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

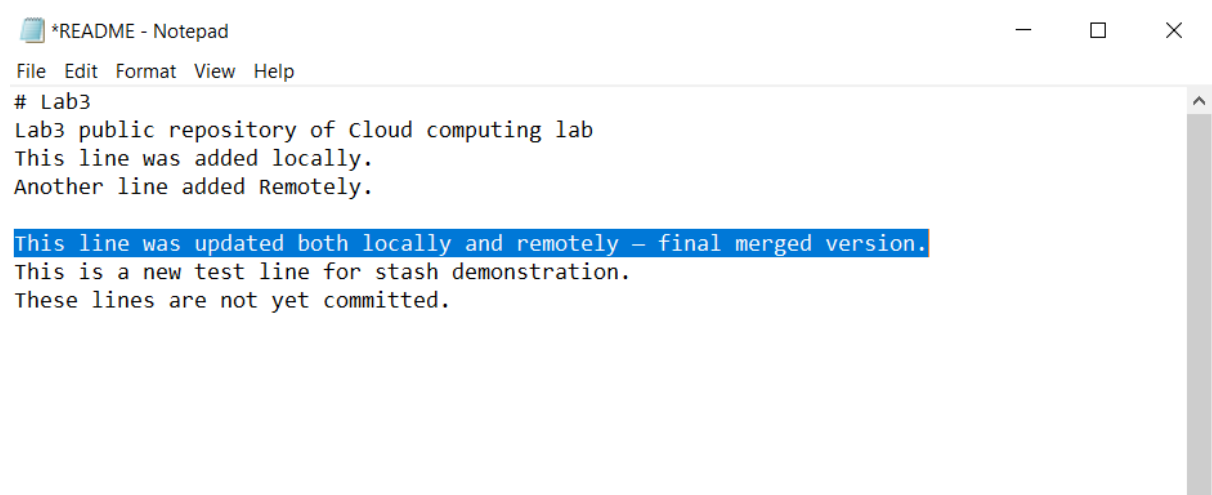
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |
```

4. Use git pull --rebase to fetch changes and trigger the conflict.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1018 bytes | 35.00 KiB/s, done.
From https://github.com/23-22411-057-sudo/Lab3
 * branch                main          -> FETCH_HEAD
  a734e00..2debad2      main          -> origin/main
Updating a734e00..2debad2
Fast-forward
 README.md | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |
```

5. Edit the conflicted file to resolve the conflict manually.



6. Mark the conflict resolved (git add <file>, git rebase --continue) and push.


```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git add README.md

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git rebase --continue
fatal: no rebase in progress

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
Everything up-to-date

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

3. Managing Ignored and Tracked Files

Steps:

1. Create a new folder (e.g., DocFiles) and add several files inside.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ mkdir DocFiles

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ cd DocFiles

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3/DocFiles (main)
$ echo "This is document 1" > file1.txt
echo "This is document 2" > file2.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3/DocFiles (main)
$ cd ..

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ ls DocFiles
file1.txt  file2.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

2. Commit and push the folder/files to GitHub.

```

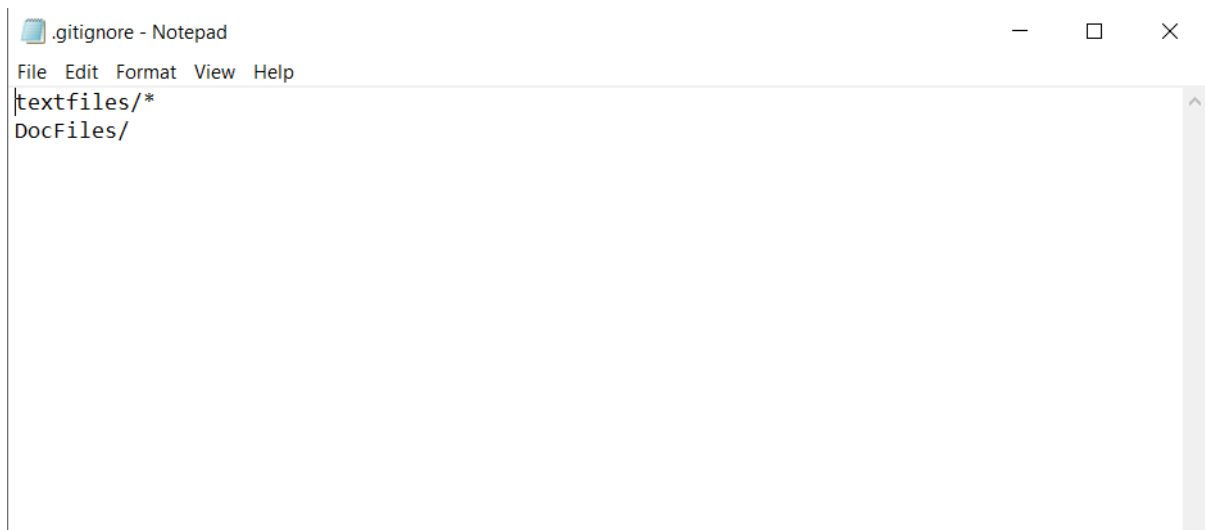
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git add DocFiles
warning: in the working copy of 'DocFiles/file1.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'DocFiles/file2.txt', LF will be replaced by CRLF the next time Git touches it

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "Add DocFiles folder with sample files"
[main 29979ea] Add DocFiles folder with sample files
3 files changed, 16 insertions(+), 3 deletions(-)
create mode 100644 DocFiles/file1.txt
create mode 100644 DocFiles/file2.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 724 bytes | 144.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
2debad2..29979ea main -> main

```

3. Add the folder to your .gitignore file.



```
.gitignore - Notepad
File Edit Format View Help
textfiles/*
DocFiles/
```

4. Commit and push the .gitignore update.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "Add DocFiles to .gitignore"
[main 4302725] Add DocFiles to .gitignore
1 file changed, 1 insertion(+)

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 508 bytes | 101.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab3.git
29979ea..4302725 main -> main
```

5. Remove the folder from tracking using git rm -r --cached <folder>.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git rm -r --cached DocFiles
rm 'DocFiles/file1.txt'
rm 'DocFiles/file2.txt'
```

6. Commit and push the change, then verify the folder is no longer tracked on GitHub.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "Stop tracking DocFiles folder"
[main b021a99] Stop tracking DocFiles folder
 2 files changed, 2 deletions(-)
 delete mode 100644 DocFiles/file1.txt
 delete mode 100644 DocFiles/file2.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 357 bytes | 119.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23-22411-057-sudo/Lab3.git
 4302725..b021a99  main -> main

```

4. Commit History Manipulation and Recovery

Steps:

1. Make a change and commit it.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ echo "Version 1 of file" > version.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git add version.txt
warning: in the working copy of 'version.txt', LF will be replaced by CRLF the next time Git touches it

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "First commit - added version 1"
[main 16185f2] First commit - added version 1
 1 file changed, 1 insertion(+)
 create mode 100644 version.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

2. Make another change and commit again.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ echo "Version 2 of file" > version.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git add version.txt
warning: in the working copy of 'version.txt', LF will be replaced by CRLF the next time Git touches it

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "Second commit - updated to version 2"
[main f66a44d] Second commit - updated to version 2
 1 file changed, 1 insertion(+), 1 deletion(-)

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

3. View your commit history.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git log --oneline
f66a44d (HEAD -> main) Second commit - updated to version 2
16185f2 First commit - added version 1
b021a99 (origin/main, origin/HEAD) Stop tracking DocFiles folder
4302725 Add DocFiles to .gitignore
29979ea Add DocFiles folder with sample files
2debad2 Remote conflict edit
a734e00 Resolved merge conflict
ad9c822 Remote edit - conflict test
1015e60 (origin/test-force) Revert "Fix log message"
625ec04 Added temporary text file
fa09b59 Fix log message
e431084 Added test line
f768264 Updated README.md on main branch
374babf (feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit

```

4. Perform a soft reset (git reset --soft HEAD~1) and observe your file and history.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git reset --soft HEAD~1

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   version.txt

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |

```

5. Make commit again.

```

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git commit -m "Third commit after soft reset"
[main 3a7127b] Third commit after soft reset
1 file changed, 1 insertion(+), 1 deletion(-)

```

6. Perform a hard reset (git reset --hard HEAD~1) and observe the changes.

```
Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git reset --hard HEAD~1
HEAD is now at 16185f2 First commit - added version 1

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ git log --oneline
16185f2 (HEAD -> main) First commit - added version 1
b021a99 (origin/main, origin/HEAD) Stop tracking DocFiles folder
4302725 Add DocFiles to .gitignore
29979ea Add DocFiles folder with sample files
2debad2 Remote conflict edit
a734e00 Resolved merge conflict
ad9c822 Remote edit - conflict test
1015e60 (origin/test-force) Revert "Fix log message"
625ec04 Added temporary text file
fa09b59 Fix log message
e431084 Added test line
f768264 Updated README.md on main branch
374babf (feature-branch) Removed tracked textfiles directory
0596303 Added .gitignore to ignore textfiles directory
0d14bfd Added textfiles directory with three files
6973dc5 Remote conflicting change
fb835f4 Second local update
4f2a732 Remote update to README
9bcde7e Initial commit

Syed@DESKTOP-S50GK51 MINGW64 ~/documents/Lab3 (main)
$ |
```